Shesh Shiromani

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EDUCATIONS

Year	Degree	Institute
Aug 2026 - Aug 2030	Bachelor's Degree (Computer Science)	Birla Institute of Technology and Science, Pilani
April 2024 - May 2026	12th Grade	Delhi Public School R.K. Puram
April 2019 - April 2024	10th Grade	Delhi Public School R.K. Puram

POSITIONS OF RESPONSIBILITY

AI Research @ Algoverse Jun 2026 - September 2026

- Selected for an intensive 4-month research cohort, mentored by senior engineers from MAANG, professors from MIT, Stanford, Caltech, UCLA and PhD Students/Researchers from reputed institutes and research labs like ETH Zurich and Scripps Research

Director, India Chapter | Edunomix Institute June 2026 - Present

- I previously served as an Econometrics Researcher and am currently building a comprehensive course aimed at discussing problem sets that contain questions from the technical rounds of Tier 1 trading firms and Investment Banking interviews. I have divided the problems into various modules like Probability, System Design, and Competitive Programming. My goal is to promote critical thinking and mathematical literacy through problem discussion.

Full Stack Developer | Inkessence June 2025 - September 2025

- As an initiative to promote early-stage authors, I was responsible for developing the Inkessence website, its publishing and purchasing system, and the machine learning model supervising its growth, from scratch.

Prompt Engineer and Front-end Development Lead | CodingEquity October 2025 - December 2025

- I was responsible for designing prompts, commercial websites, and the management of the front-end development team. At CodingEquity, my goal was to come up with the most innovative graphic design solutions and implement them using web development and prompt-engineering skills.

Director of Research and Development | Kronos May 2026 - August 2026

- We were developing and training a supervised machine learning model using datasets from various aerospace agencies like NASA, SpaceX, and ISRO to simulate planetary conditions in extraterrestrial environments. Once the prototype was ready, we planned to deploy it using AR/VR consoles and subsequently develop a remote GUI for the general public to access. We also collaborated with ISEF Awardees and IESO Medalists to get idea validation and further guidance on this project.
- I was also responsible for the merit-based selection of all core team members and departmental leads by conducting various technical and non-technical interview rounds.

Student Researcher | The New York Academy of Sciences August 2025 - October 2025

- I contributed to Heliogrid, a team project modelling solar irradiance dynamics and their potential impact on atmospheric ozone concentration. My role involved the literature review of atmospheric physics papers, contributing to the team's computational modelling framework, and co-authoring the research proposal. I previously reviewed and evaluated expository physics papers submitted by fellow Junior Academy members across thermodynamics and classical mechanics.

CLUBS

President of PhySoc | Physics Society of DPS RK Puram July 2023 - May 2026

- I was leading the Physics Society with 130+ members. My responsibilities included overall club management, organising the annual PhySoc inductions across all departments, organising the annual event Physium, managing the drafting and publication process of the annual magazine Phyzard, and managing a team of 6 departmental heads. I was also mentoring junior members in research and question paper design to participate in various physics competitions such as OPhO and Physics Cup. Under my tenure, departmental expansion of PhySoc was held, and its research output has significantly increased.

Previously served as the Departmental Head of Research and Question Paper Design departments, where I was responsible for:

1. In charge of the question paper design process for the PhySoc Common Paper.
2. Designed research, design, and group discussion prompts for their respective departmental inductions.
3. Reviewed various expository research papers written by participants and selectively filtered out the qualifying ones.
4. Responsible for entire departmental management, along with a few other senior members.
5. Conducted 50+ interviews for selectively assessing the skills of current PhySoc Members.

Senior Member @ CyberX | Cybersecurity Club of DPS RK Puram July 2025 - May 2026

- Wrote a research paper titled "Unlocking the Mathematical Code Behind Cybersecurity", which was selected for the CBSE Cyber Surakshit Bharat Competition and was also published on the Resources page of the CyberX website.
- Took part in Cryptic Hunts and CTF events, placing 50th (twice) in Sudocrypt (annual cryptic hunt of DPS RK Puram, organized by Exun) out of 300+ participants.
- Proctored the Inter-DPS Quiz Finals.

Senior Member @ Digex | Photography and Filmmaking Club of DPS RK Puram August 2025 - May 2026

- I mainly worked with the technical department, where I was responsible for designing the back-end framework for the Digex website alongside fellow team members, as well as some ground-level volunteering for annual events and competitions.

Core Tabla Player and Beatboxer | Indian Instrumental Club April 2019 - May 2026

Under the guidance of Mr. Kuldeep Mishra.

- Lead Tabla player for annual SUPW showcase, followed by a pair-wise beatboxing performance.
- Participated in the instrumental team for a semi-core fusion song, which was played at the opening ceremonies of the Class 12th Scholar Badge ceremony and Day 1 of Commastitva Event (annual commerce fest).

Member of the National Swimming Team | DPS Noida and DPS RK Puram

- Selected for CBSE National Swimming Meet (Delhi side), got 4th position in butterfly and relay finals.
- Represented Delhi at Khelo India Swimming Meet (state level), got 4th position in freestyle and relay finals.

SUMMER PROGRAMS / VIRTUAL TRAININGS

- IBM Qiskit Global Summer School (Free)
- Atlas Financial Literacy Summer Program (Free)
- GLT Virtual Summer Program (Girls Leading Tech) (Free)
- Restless Dreamers Summer Fellowship (100% Scholarship)
- International Summer School, New Delhi(100% Scholarship)
- Harvard Ventures TECH Summer Program
- Inspirit AI Scholar's Program
- IMUN Campus Ambassador Internship
- YRI Research Fellowship

AWARDS & ACHIEVEMENTS

1. Principal's Gold Medal - 7 consecutive years of academic excellence
2. Principal's Blue Tie - 6 consecutive years of academic excellence
3. Principal's Blue Blazer - 5 consecutive years of academic excellence
4. Principal's Scholar Badge
5. Got Class Rank 2 in Mathematics and overall aggregate rankings - 10th Grade AISSE
6. Received a 100% Scholarship in the Big Bang Edge Scholarship Examination - FIITJEE Entrance Test
7. Received high honours in the Online Physics Olympiad (OPhO) - Physoly and PhODS
8. Selected for IICPC Summer Trading Project
9. Won YRI Fellowship Abstract Writing Competition - Received 50% Scholarship for research fellowship
10. Qualified for Semi-Finals of the Global Quantum Mechanics Challenge (Received Special Honours)
11. Qualified for the semi-finals of International Youth Math Challenge(IYMC)
12. Qualified for the semi-final round of the International Computer Science Competition
13. Contributed to MIT PRIMES CrowdMath
14. Achieved an overall Rank of 7 out of 122 Teams in BrainRot'26 (Maths and Logical Puzzles College Competition) - SST

PROJECTS

BenchRank - Low Latency Trading Infrastructure

- A distributed benchmarking and hosting platform engineered for evaluating high-performance trading infrastructure under extreme concurrent load.
- Built as part of the IICPC Summer Hackathon 2026, the system enables contestants to deploy custom matching engines and exchange backends inside isolated containerized environments, after which a distributed bot fleet stress-tests the infrastructure with large-scale simulated market activity.
- The platform measures real-time latency distributions, throughput ceilings, execution correctness, and system resilience while streaming live telemetry and rankings through a dynamic leaderboard interface.
- The project explores systems engineering at the intersection of distributed computing, low-latency infrastructure, container orchestration, telemetry ingestion, and concurrent network simulation.
- Tech stack included containerized deployment pipelines, real-time analytics streaming, scalable load generation architecture, and fault-isolated execution environments.

Rasoi Saathi - Smart Kitchen Assistant

- I built a smart kitchen assistant, purely using LLMs, for Wispr Flow x Lovable's AI Hackathon. It allows the user to design menus, choose meals, and choose recipes from a database of 200+ traditional Indian recipes. With customization available at every point, meal preferences and allergies can also be added.

Nomex - IUPAC Name Generator

- Built the backend using Python and the frontend using React with basic HTML and CSS. The application accepts compound details or a common name as input and returns the corresponding IUPAC name through the prototype generation pipeline.
- (Only a prototype; frontend still in production)

ribock - Personalized Roadmap Generator for Competitive Programmers

- Currently in production. The platform generates individualized competitive programming roadmaps from a user's current rating on recognized CP platforms and maps progression against a selected objective, including ICPC preparation, interview preparation, or general problem-solving.
- The system structures the roadmap as a goal-conditioned progression layer that translates current skill/rating signals into an ordered set of practice targets and learning milestones.

terra - Personalized AI Assistant

- Built a lightweight desktop AI assistant using Python, with pyttsx3 for text-to-speech and smtplib for email automation. The system integrates voice-oriented interaction with utility and automation workflows to execute routine actions without requiring direct computer interaction.
- Designed as a compact personal assistant architecture focused on combining conversational output, speech synthesis, and automated communication into a single local workflow.

RESEARCH PUBLICATIONS (EXPOSITORY)

Beyond Constant Volatility: Integrating Jump-Diffusion and Advanced Integration Techniques into the Black-Scholes Framework for an Option Pricing Model

- Authored a comprehensive research proposal for the YEL Student Scholars Contest (in partnership with Synthica) focused on advancing the predictive accuracy of the Black-Scholes-Merton model. This project addresses the limitations of the standard model by integrating Merton's Jump-Diffusion and stochastic volatility to better capture market phenomena like volatility smiles and discontinuous price trajectories.

You can feel the heat, but can you measure it?

- This expository research paper develops a precise thermodynamic distinction between heat and temperature using kinetic theory, thermal conductivity variation, and energy exchange rate analysis. It demonstrates why objects at equal temperatures produce different thermal sensations.

Unlocking the Mathematical Code Behind Cybersecurity

- Shortlisted for submission to the CBSE National Level Cyber-Security Awareness Competition.
- I had written an expository research paper on demystifying some mathematical foundations that are frequently used in Cybersecurity. This publication explores how hashing and encryption utilize concepts from permutations and combinations, linear algebra, matrix multiplication, and other related fields.

Unlocking the Mysteries of Interstellar Time Travel

- This paper presents a conceptual analysis of relativistic and thermodynamic constraints on interstellar travel timescales, drawing on Kip Thorne's work on closed timelike curves and Nolan's Interstellar as a starting framework.

SKILLS

Programming Languages : Python, C++, Java, JavaScript, Lua, QBASIC, Ruby, SQL
Libraries / Frameworks: NumPy, Pandas, Matplotlib, Flask, Django, PyTorch, TensorFlow
Coursework: Probability, Discrete Mathematics, Data Structures and Algorithms, Web Development